

A Are US-matched firms different from non-matched firms?

While the EIBIS results indicate virtually no aggregate difference in AI adoption between US and EU firms, particularly in financially developed countries (see Section 3.1), it is important exercise to examine which US firms are selected by the matching algorithm and how they differ from their non-matched counterparts, for at least three reasons.

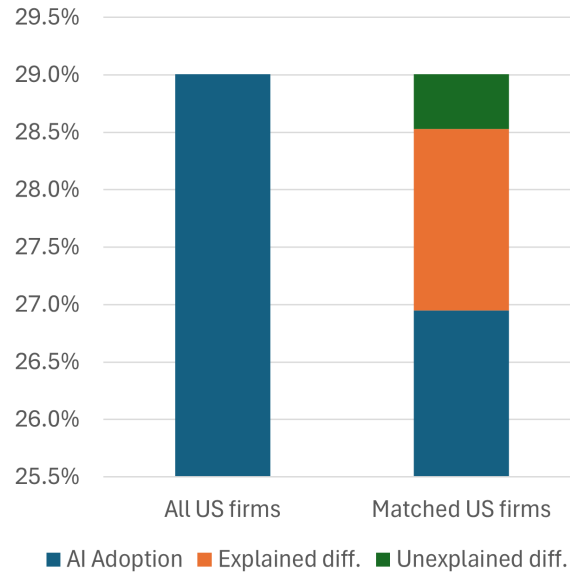
First, the comparison can provide a robustness check on the matching methodology itself, shedding light on whether the algorithm systematically favors certain types of firms (e.g. larger, more innovative or less financially constrained). Second, even if average adoption rates are similar across the US and EU, the characteristics of firms driving adoption may diverge, with potential implications for productivity dynamics. Finally, contrasting matched and non-matched US firms can reveal whether transatlantic comparisons are shaped by differences in firm composition rather than adoption per se, thus helping to interpret the aggregate similarities more carefully.

To this end, we apply the Oaxaca–Blinder decomposition to a synthetic dataset that compares all US firms with the subset of US firms identified as nearest neighbours in the matching procedure. In the exercise, we include only firms that disclose the full set of matching variables such that the average AI adoption rate in the sample deviates slightly from the unconditional US average. We therefore treat this as an approximation.

The decomposition, originally developed by [Blinder \(1973\)](#) and [Oaxaca \(1973\)](#), partitions the mean difference in AI adoption into components explained by observable firm characteristics and a residual unexplained part. Consistent with the matching strategy described in Section 4, we compute the decomposition relative to sector- and size-normalised means of value added, which controls for compositional shifts in the fixed effects.

Figure A.1 shows that US-matched firms exhibit, on average, a 2 percentage points lower AI adoption rate than the full set of US firms. Close to 80% of this gap is explained by differences in matching variables and sector- and size-class composition. This supports the robustness of the matching design.

Figure A.1: AI adoption among US-matched and all US firms



Note: Average AI adoption rates across all US firms and US firms matched with EU counterparts. Propensity score matching is based on the 5 nearest neighbors in terms of investment intensity, innovator binary indicator, bond issuance binary indicator and business monitoring system binary indicator in the same sector and company size category. The difference between the unconditional and conditional US means is decomposed into the component that can be explained by the matching variables, normalization fixed effects and the residual, based on the Oaxaca-Blinder decomposition. Values are weighted by value added.

B Additional results

Table B.1: Explanatory power of matching variables for AI adoption

	(1) AI Adoption EU	(2) AI Adoption US
Investment/employee (log)	0.017*** (0.001)	0.014*** (0.004)
Business monitoring system	0.111*** (0.004)	0.124*** (0.014)
Equity/bond issuance	0.092*** (0.018)	0.059 (0.048)
Innovator	0.058*** (0.004)	0.050*** (0.013)
Size FE	Yes	Yes
Age FE	Yes	Yes
Country-sector-wave FE	Yes	Yes
Observations	45785	2966
R-squared	0.200	0.124
Adjusted R-squared	0.168	0.102

Note: The dependent variable is AI adoption and is derived from EIBIS. Standard errors clustered at the country-sector-wave level are reported in parentheses. *, **, *** correspond to 0.1, 0.05 and 0.01 significance levels, respectively.

Table B.2: Effects of AI adoption on labor productivity with different fixed effects

	(1) LP (log) With controls	(2) LP (log) With controls	(3) LP (log) With controls	(4) LP (log) With controls	(5) LP (log) With controls
AI adoption (US-based)	0.076*** (0.015)	0.033** (0.015)	0.040*** (0.011)	0.042*** (0.013)	0.041*** (0.011)
Investment (lag)	0.008 (0.005)	0.009 (0.006)	0.012** (0.005)	0.010* (0.005)	0.011* (0.005)
Profitability (lag)	0.694*** (0.058)	0.591*** (0.047)	0.657*** (0.050)	0.661*** (0.045)	0.661*** (0.049)
Financial leverage (lag)	-0.128*** (0.046)	-0.188*** (0.031)	-0.131*** (0.036)	-0.132*** (0.026)	-0.131*** (0.036)
Total assets (log, lag)	0.471*** (0.012)	0.408*** (0.007)	0.389*** (0.012)	0.387*** (0.006)	0.388*** (0.012)
Size FE	Yes	Yes	Yes	Yes	Yes
Age FE	Yes	Yes	Yes	Yes	Yes
Sector FE	No	No	No	Yes	No
Country FE	No	No	Yes	No	No
Sector-wave FE	Yes	No	Yes	No	Yes
Country-wave FE	No	Yes	No	Yes	Yes
Observations	30327	30657	30327	30327	30327
R-squared	0.432	0.441	0.513	0.516	0.517
Adjusted R-squared	0.430	0.437	0.511	0.513	0.514

Note: The dependent variable is labor productivity, calculated as the log of turnover per employee, and is derived from EIBIS. AI adoption (US-matched) is the AI implementation status derived from similar firms in the US. Investment is expressed as the annual change in total fixed assets. Profitability is the ratio of EBIT to total assets. Financial leverage is the ratio of loans and long-term debt to total assets. All control variables come from ORBIS and are lagged by 1 year. Standard errors clustered at the country-sector-wave level are reported in parentheses. *, **, *** correspond to 0.1, 0.05 and 0.01 significance levels, respectively.